Does increasing complexity change behaviour under ambiguity? "Less is More"

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Abstract

This paper challenges the conventional view that complex decisions under ambiguity require complex models. We hypothesize instead that increasing complexity causes cognitive overload, making choices noisier and simpler models more robust. Results confirm that 'less is more'. Under ambiguity and complexity, individuals simplify their decision process, defaulting to risk-like heuristics. An allocation method is used in the experiment, where complexity was controlled across ambiguous lotteries by varying the number of outcomes and stages. Then we competitively evaluate the Expected Utility (EU), αMaxMin (AMM), and Choquet Expected Utility (CEU) models. As complexity rises, the EU model provides the best fit, while the more complex AMM and CEU models overfit the increased decision noise. The EU model's risk-attitude parameter remains stable, whereas the additional parameters in AMM and CEU become unreliable under cognitive strain. The CEU model performs well only after participants familiarise problems with less noise, allowing its ambiguity parameter to function. Our finding implies that in complex real-world settings accurate prediction may depend more on relatively simple models that reflect this cognitive reality than on theoretically sophisticated alternatives.

JEL codes: D81, D91, C91

Keywords: Ambiguity, Complexity, Allocation Elicitation Method, Model Fitting, Experiments

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1. Introduction

Consider an investor allocating a retirement portfolio in a volatile market. They must weigh familiar risks, where outcomes have known probabilities, like the historical volatility of a stock index, against deeper ambiguities, where the probabilities are fundamentally unknown. Is a new geopolitical conflict 10% likely or 50%? Will a breakthrough in technology succeed in 3 years or 10? Facing this complex mix of risky and ambiguous prospects, does the investor employ a sophisticated mental calculus, taking into account this 'unknown-ness'? Or, overwhelmed by the complexity, do investors simplify, perhaps treat all ambiguous prospects as if they were merely risky? Such a scenario highlights a critical distinction in decision theory between risk and ambiguity. It also frames my core research question: As choice problems become more complex, do we need more complex models to describe behaviour under ambiguity, or do individuals cognitively simplify, making relatively simple models more accurate? This question urgently needs to be investigated because the real world is replete with ambiguous, multi-stage problems, yet most models are tested in simpler, one-stage settings.

To explore the question above discussed, a laboratory experiment was conducted. Using the allocation method, we examine how individuals distribute endowments across progressively complex ambiguous lotteries, where only the lowest and highest probabilities of possible outcomes were informed to them. Complexity is divided into two dimensions: number of outcomes and number of stages. Three models: the simple Expected Utility (EU) model, which does not explicitly model ambiguity attitudes, and the more complex αMaxMin (AMM) and Choquet Expected Utility (CEU) models are applied to the data. By checking the goodness-of-fit, we can capture individual behaviour patterns as complexity increases under ambiguous conditions and hence determine how subjects are tackling the problems.

Counter to the intuitive notion, that complex ambiguous problems demand equally complex models, our results demonstrate that 'less is more'. This finding can be explained by the cognitive load which indicates the mental effort imposed on working memory when processing information (Sweller, 1988). In decision-making under ambiguity and structural complexity, expanding possible outcome space or multiple stages increases cognitive load by requiring individuals to simultaneously evaluate more information and integrate uncertainty across levels. When cognitive load exceeds processing capacity, individuals may abandon sophisticated ambiguity calculus and adopt simplifying heuristics, such as treating unknown probabilities as if they were known. From a model-fitting perspective, this behavioral simplification manifests as noisier choices, causing the complex AMM and CEU models to overfit while the EU model provides the most robust account of behaviour. This finding offers a suggestion for policymakers and product designers that in complex real-world environments from financial markets to healthcare choices, predicting behaviour requires models that account for this cognitive simplification, not those that assume sophisticated ambiguity calculus.

2. Related Literature

The Ellsberg Paradox (Ellsberg, 1961) revealed a robust deviation from Expected Utility Theory (Savage, 1954). It showed that individuals consistently prefer bets with known probabilities over bets with unknown probabilities, even when the objective odds are identical. This finding triggered a wide series of studies on ambiguity.

Initial explanations focused on psychological heuristics and pessimism. The breakthrough came with formal models that could incorporate ambiguity attitudes. Choquet Expected Utility (Schmeidler, 1989) and MaxMin Expected (Gilboa & Schmeidler, 1989) introduced the notion of multiple priors, where individuals were assumed to consider a set of possible probability distributions. The former challenged additive probabilities (necessary under Expected Utility theory) in ambiguous situations and captured individual ambiguity attitudes via the non-additivity. The latter proposed that for those people who are ambiguity averse, they evaluate choices based on the worst-case scenario. Then, the αMaxMin model (Ghirardato et al., 2004) and the Smooth Model of Ambiguity Attitudes (Klibanoff et al., 2005) separated ambiguity perception from ambiguity attitudes, allowing for the empirical measurement of an individual's ambiguity aversion coefficient, analogous to risk aversion coefficients.

While some research attributes the reduction of ambiguity aversion to perceived competence (Heath & Tversky, 1991) or the use of frequentist mappings (Fox & Tversky, 1995), a parallel literature in risk perception has focused on a different moderating variable: complexity. The relevant studies can be grouped by two characteristics of a risky situation: the number of possible outcomes and the number of stages. Mowen and Gentry (1980) and subsequent studies show individuals prefer simple 50-50 gambles over equivalent multi-outcome gambles with the same Expected Value. One possible explanation for this behaviour pattern is that a broader outcome distribution makes the potential for losses more salient or increases regret anticipation. As for the latter, Decision-makers seem to fail to reduce compound (or multi-stage) lotteries. Instead, they evaluate stages in isolation (narrow framing), leading to time-inconsistent risk attitudes. One behavioural resolution is that people evaluate each stage of a multi-stage risk separately, not as an integrated whole. An experimental evidence (Halevy, 2008) shows that people are more risk-averse in a two-stage risky lottery where the first stage determines the probability for the second, compared to its equivalent reduced simple form.

While the literatures on ambiguity attitudes and decision complexity have evolved in parallel, their intersection remains sparse and underexplored. This is a significant gap: the canonical experiments like the two-urn Ellsberg paradox establishing ambiguity aversion intentionally use a simplified design with two outcomes and a single stage. Consequently, little is known about how the structure of a decision problem affect individual behaviours when probabilities are unknown.

The limited research on the relationship of ambiguity attitudes with complexity suggests that complexity is not a neutral background variable, but an active moderator of ambiguity attitudes. Based on the three-colour Ellsberg urns test, Chew et al. (2017) found that the choice depends on the ranking of the ambiguous event within the outcome distribution, suggesting that the valuation of ambiguity is varied cross all payoffs in a multi-outcome setting. The work by Abdellaoui et al. (2011) on measuring ambiguity attitudes through multiple pairwise comparisons also implicitly dealt with a richer outcome space, noting the challenge of stabilizing measurements, which may itself be a symptom of complexity-sensitive attitudes. Based on these findings, expanding the number of ambiguous outcomes may affect individuals' attitudes toward ambiguity but in which way still needs to be further explored.

As for the studies related to the dynamic complexity under ambiguity situations, Halevy (2008) proposed a crucial finding that aversion to compound (two-stage) ambiguity is significantly stronger than aversion to one-stage ambiguity or compound risk. This demonstrates a violation of the reduction principle for ambiguous lotteries and suggests that the process of uncertainty resolution generates a unique and potent form of aversion, especially when intermediate stages are also ambiguous. However, Halevy's design primarily establishes the existence of this effect using a minimal paradigm. My research systematically varies the degree of complexity.

This project moves beyond demonstrating that complexity can affect individuals' decision patterns under ambiguity context, to a systematic investigation of how and why these behaviours change as a function of two well-defined complexity dimensions: outcome space complexity and dynamic complexity. Unlike most of previous research, this study considers the combination of these two dimensions to further investigate the factor which may affect the aversion most. Also, by comparing model fitness for different kinds of lotteries in the experiment, it captures how decision-makers evaluate choices dynamically. It therefore makes a novel contribution by systematically dissecting the cognitive architecture of ambiguity attitudes, moving beyond static models to reveal how decisions are dynamically constructed by multiple structures of problems.

3. Experimental Design

There were six sessions with 100 participants in total from the University of York. On average, each session lasted around 1 hour, and participants received £16.83 pp including £2.50 show-up fee. The total payment of the experiment was £1661.01[1]. During the experiment, participants were asked to answer a series of allocation problem over four parts: Part RL, Part ALA, Part ALB and Part ALC. As this experiment mainly applied the allocation method, it is necessary to start with an explanation of how it is used to elicit preferences.

3.1 Allocation method

Participants are offered a certain number of tokens, F, as their endowment, and are asked to allocate it between two options with exchange rates between money and tokens. To demonstrate its application in preference elicitation, one of allocation problems in Part RL will be explained in detail below. Options will be referred as colours.

For this problem, participants need to decide how to allocate $F = 100$ tokens between Red and Yellow. The probability of Red happening is p which is 70%, and that of Yellow happening is $1\text{-}p = 30\%$. The exchange rate of Red, E_1, is £0.06 so one token allocated to Red is worth 6 pence, and one token allocated to Yellow is worth 8 pence (i.e. $E_2 = £0.08$). If this problem is chosen for determining participants' payments for this part, the computer will choose either Red or Yellow. Suppose a participant allocates X tokens to Red: she/he will get £XE_1 if Red is played out by the computer. Otherwise, she/he will receive £$(F\text{-}X)E_2$. Figure 1 displays the software interface of this problem. In this case, $X = 50$, so the participant can get the amount of money allocated to Red, £3 = 50 × 0.06, with chance 70% and get £4 which is the amount of money allocated to Yellow with chance 30%. Participants can choose their allocation by moving the slider. It should be noted that changing the allocation only affects the amounts of money allocated to Red and Yellow but not probabilities and exchange rates. The right-hand side histogram in Figure 1 visualizes the allocation. The horizontal axis shows the chances, and the vertical axis shows the amounts of money, so the chance of Red being chosen is fixed and shown by the horizontal distance of the Red rectangle – in this case 70%. Similarly, the chance of getting the money value of the tokens

[1] I apricate Prof. John Hey's kind individual funding for this experiment.

allocated to Yellow is 30%. Moving the slider will change heights of the rectangles change solely, but not their width. Probabilities of Red happening and Yellow happening, and exchange rates of them only change between problems. An allocation indicates a mixture of Red and Yellow. Participants can control how much risk they would expose to by changing the allocation. For example, an extreme risk-averse person may follow this half-allocation strategy to guarantee she/he can always get positive payments. If this problem were to be chosen by the computer at the end of Part RL, the lottery in Figure 1 would be played out. The spinner would be spun by the computer, and where it stops determines the participant's payoff for this problem (Figure 2).

Figure 1. An example of a problem in the Risk part with an equal split allocation.

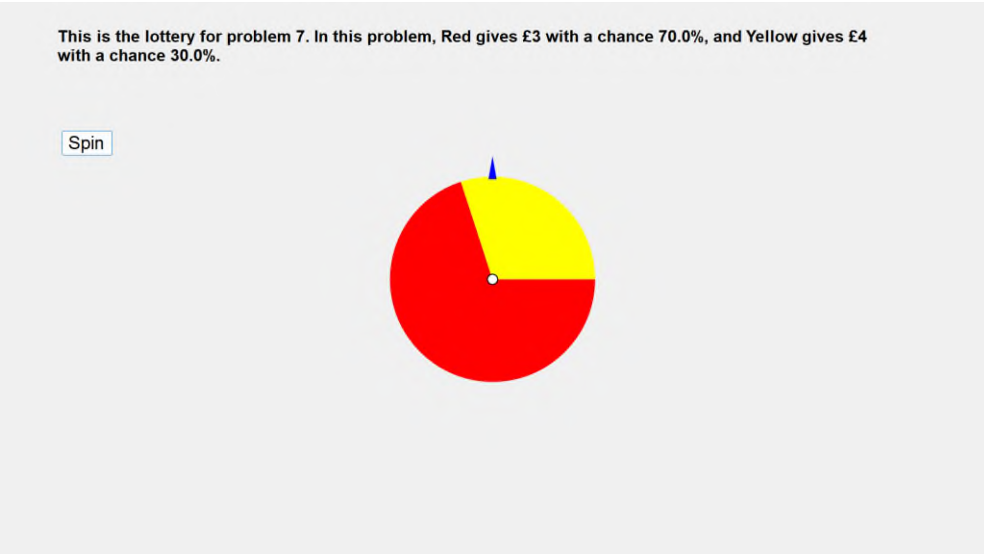

Figure 2. The computer choosing the problem

3.2 The Experiment

The experiment is in four parts: Part RL; Part ALA; Part ALB; Part ALC. In each part there are several allocation problems: 10 in Part RL; 10 in Part ALA; 15 in Part ALB; and 15 in Part ALC. After each part, the computer chooses a problem at random, and the payoff on that problem is the participant' payment for that part.

Overall, Part RL contains the simplest risk allocation problems, and the remaining three parts involve ambiguity which is reflected in the lack of knowledge of probabilities. **Participants are informed the lowest probability and the highest probability of a colour happening.** The complexity of problems increases over parts from ALA to ALC in terms of two dimensions: the number of possible outcomes and the number of stages. These four parts are discussed below.

i. Part RL (Risky Lottery)

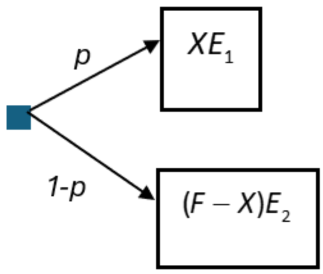

Figure 3. Game tree of questions in Part RL

To summarise, this is a one-stage risky lottery. A decision-maker (DM) is offered the endowment F and asked to decide how many tokens allocated to Red X with the exchange rate E_1, and Yellow $(F\text{-}X)$ with the exchange rate E_2. The probability of Red happening is p and that of Yellow happening is $1\text{-}p$. If Red happens, the DM will get XE_1. If Yellow happens, the DM will get$(F\text{-}X)E_2$. Figure 1 has shown the software interface of this part.

ii. Part ALA (Ambiguous Lottery A)

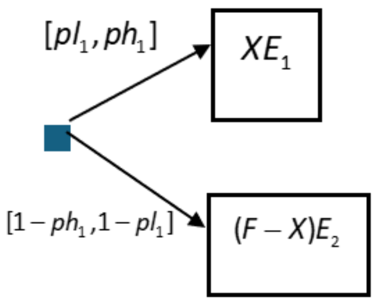

Figure 4. Game tree of questions in Part ALA

This is a one-stage ambiguous lottery. A DM is offered the endowment *F* and asked to decide how many tokens allocated to Red *X* with the exchange rate E_1, and Yellow *(F-X)* with the exchange rate E_2. The DM is only told that the highest probability of Red happening is ph_1, and the lowest probability of Red happening is pl_1. Therefore, the highest probability of Yellow happening is $1\text{-}pl_1$ and the lowest probability of Yellow happening is $1\text{-}ph_1$. If Red happens, the DM will get XE_1. If Yellow happens, the DM will get $(F\text{-}X)E_2$.

iii. Part ALB (Ambiguous Lottery B)

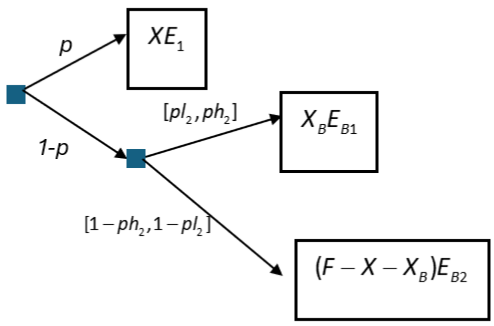

Figure 5. Game tree of questions in Part ALB

This is a **two-stage** ambiguous lottery. A DM is given an endowment of *F* tokens to allocate between Red, with probability of Red happening *p*, and another event which has the residual probability of *1-p* of happening. Denote the allocation to Red by *X*. There is an exchange rate E_1 between tokens and money for Red. If Red happens, then the DM will get paid XE_1. If Red does not happen, the DM will have to take another allocation decision: out of the remainder of the endowment *(F-X)*, the DM must choose X_B to allocate to Blue which has the highest probability ph_2 of occurring and the lowest probability pl_2 of occurring - if it does occur the DM will get paid $X_B E_{B1}$; where E_{B1} is the exchange rate between tokens and money for Blue; if it does not occur, the DM will get paid $(F\text{-}X\text{-}X_B)E_{B2}$ where E_{B2} is the exchange rate between tokens and money for Purple.

iv. Part ALC (Ambiguous Lottery C)

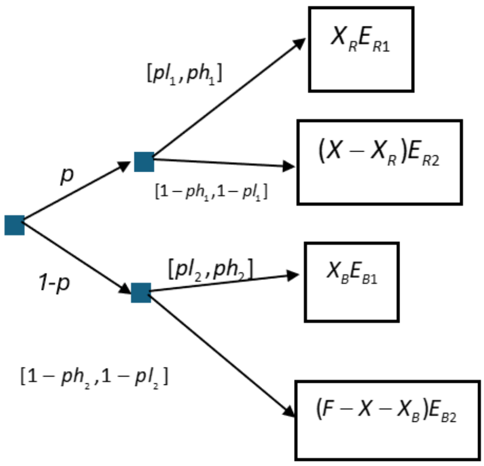

Figure 6. Game tree of questions in Part ALC

This is a **two-stage** lottery with **two** ambiguous branches. A DM is given an endowment of *F* tokens to allocate between the event (Red or Orange) with probability *p* of happening, and another event (Blue or Purple) which has the residual probability of *1-p* of happening. Denote the allocation to Red or Orange by *X*. If Red or Orange does happen, the DM is asked to take another allocation decision: based on the endowment *X* to choose X_R to allocate to Red which has the highest probability ph_1 of occurring and the lowest probability pl_1 of occurring with the exchange rate E_{R1}. If Red happens, the DM will get $X_R E_{R1}$. If Red does not happen, the DM will get paid $(X-X_R)E_{R2}$, where E_{R2} is the exchange rate between tokens and money for Orange. Similarly, If Red or Orange does not happen that is, (Blue or Purple does happen), the DM is asked to take another allocation decision: out of the remainder of the endowment *(F-X)*, the DM must choose X_B to allocate to Blue which has the highest probability ph_2 of occurring and the lowest probability pl_2 of occurring – if it does occur the DM gets paid $X_B E_{B1}$ where E_{B1} is the exchange rate between tokens and money for Blue; if it does not occur, the DM gets paid $(F-X-X_B)E_{B2}$ where E_{B2} is the exchange rate between tokens and money for Purple.

The software interfaces of these three ambiguous lottery parts are displayed in Figure 7, 8 and 9 respectively.

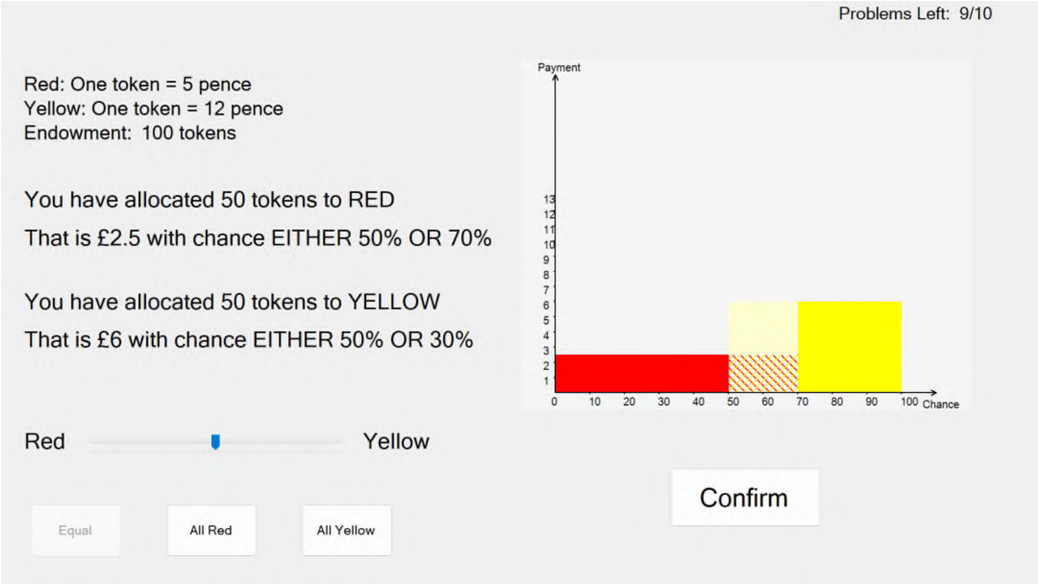

Figure 7. Software interface of Part ALA

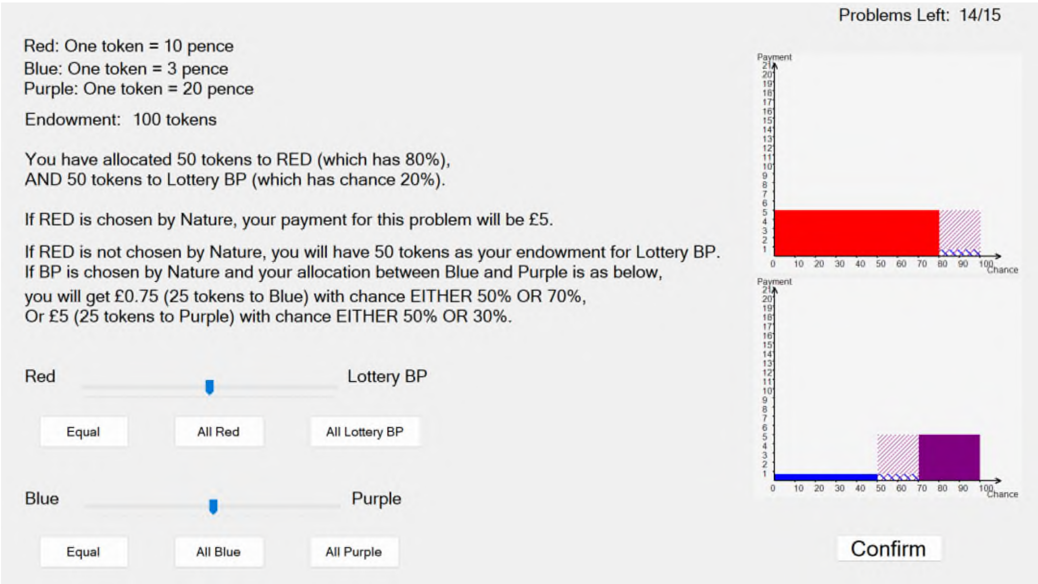

Figure 8. Software interface of Part ALB

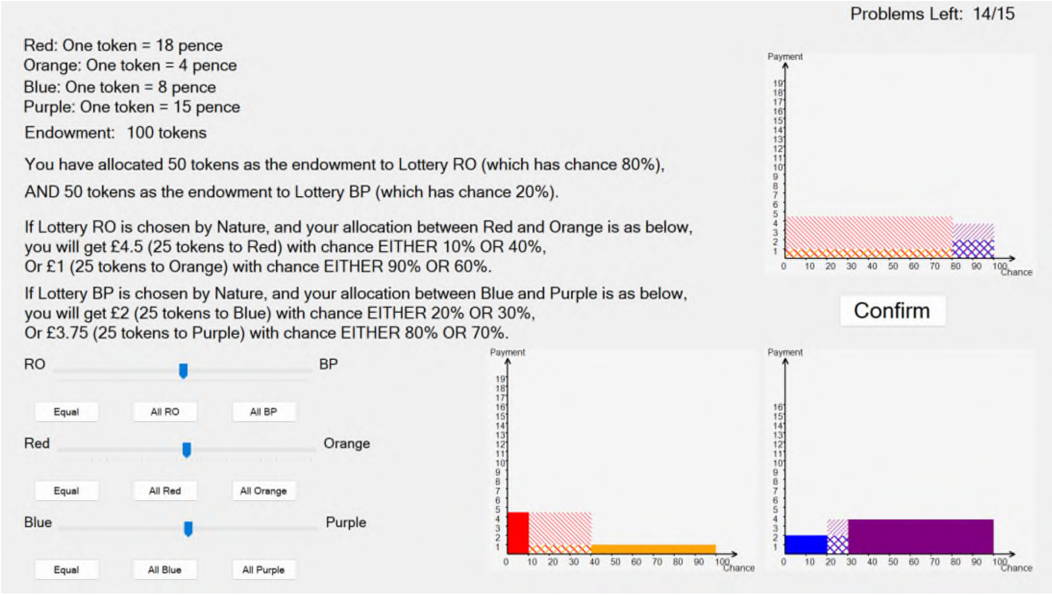

Figure 9. Software interface of Part ALC

4. Theoretical Framework

The experiment is designed to elicit individuals' attitudes towards risk and ambiguity as we want to investigate as the complexity of ambiguous lotteries increases, whether their attitudes towards ambiguity change or not. To estimate individuals' attitudes to ambiguity, as we are going to estimate their parameters we need to know their actual allocation decisions, as well as their optimal allocations. These latter depend on the ambiguity model that we use. We are going to assume that these latter are: (Subjective) Expected Utility ((S)EU), αMaxMin (AMM) model and Choquet Expected Utility (CEU). The theoretical optimal allocation *X* for each problem in the experiment can be calculated based on these models. As the data clearly contains a random component, we model this using the familiar Stochastic Decision Rule (McFadden, 1972), the actual allocation (*AA*) is assumed to equal the optimal allocation (*X*) plus noise with a normal distribution $(0,\sigma^2)$ in each problem. Such noise would change over parts and be different among individuals: $AA_{ijk} = X_{ijk} + \varepsilon_{ik}$, where $\varepsilon_{ik} \sim (0,\sigma_{ik}^2)$, *i* indicates the subject number $i = \{1,2,\ldots,100\}$, *j* represents the problem number within each part and *k* here indicates the part $k = \{RL, ALA, ALB, ALC\}$. The optimal allocation is a function of key parameters from these models with given experiment parameters - probabilities and exchange rates – as I discuss below.

The methodology is as follows:

We estimate subject by subject, part by part and model by model. For each subject, each part and each model, we take the actual allocations and estimate the parameters of that model by maximum likelihood. We have to proceed sequentially to overcome identification problems. We start with Part RL and use the data to estimate *r* the coefficient of risk aversion. We then, using this and the data from Part ALA, estimate the parameter α*,* of the AMM model and γ for CEU. Finally, I use the data from Part ALB and estimate the parameter γ *o*f the CEU model.

An inevitable by-product of the maximum likelihood estimation is, for each subject, each part and each model, the value of the maximised log-likelihood. This is an obvious measure of the goodness-of-fit of that model for that subject for that part and that model. To compare these goodness-of-fit across models and parts we need to work with a corrected Akaike Criterion considering differences in the number of parameters and the number of observations.

4.1 Model

The underlying utility function is assumed to be the CRRA model: $u(x,r)=\frac{x^{1-r}}{1-r}$ for $r\geq 0$ and $r\neq 1$; $u(x)=\ln(x)$ for $r=1$. For the former, as a more general case, *r* is the CRRA index of risk aversion and *u* takes the form $u(x,r)=\frac{x^{1-r}}{1-r}$ and $u'(x,r)=x^{-r}$. For the latter, $u(x)=\ln(x)$ and $u'(x)=x^{-1}$. To find the optimal allocation, I took the first-order derivative of the expected utility with respect to *X* and put the answer equal to 0.

a. (S)EU model

It is the standard expected utility theory with the key parameter *r* from the CRRA model. If $r=0$, individuals would be risk neutral. If $r>0$, they would be risk averse, and the larger the *r* is, the more risk averse the DM. Otherwise, they are risk seeking In an ambiguous situation, it becomes the subjective EU model which assumes that individuals make decisions based on their *subjective* beliefs about the probabilities (Savage, 1954). As the target is to estimate *r* and check if *r* changes over three parts (Part ALA, Part ALB and Part ALC), the subjective probability in the ambiguous case was simply assumed to be the average of the highest probability, *ph*, and the lowest probability, *pl*.

b. AMM model

This model focuses on individual decision making under ambiguity. Developing from the MaxMin model (Gilboa & Schmeidler, 1989), it introduces a new parameter α to generalize this story (Ghirardato et al., 2004). The AMM model assumes that people consider both the best case and worst case in a weighted way. $\alpha \in [0,1]$ is not only the weighted average parameter but also the ambiguity aversion index. If $\alpha = 1$, people would be fully pessimistic, only looking at worst cases, and the AMM model reduces to the MaxMin model. In contrast, if $\alpha = 0$, only best cases matter, and the model reduces to the MaxMax model. When $\alpha = 0.5$, individuals weight these two extreme situations equally. Combining the setting of the experiment and the assumption in the EU model, under such situation, the AMM model reduces to the EU model mathematically. Individuals are treated as ambiguity neutral, as they behave similarly to a standard EU maximiser with a 'representative' prior $p = \frac{pl + ph}{2}$. Generally, if $0 \le \alpha < 0.5$, people would be ambiguity seeking; if $0.5 < \alpha \le 1$, they would be ambiguity averse.

c. CEU model

The standard EU model assumes that individuals make decisions based on additive assigned probabilities: $P(A \cup B) = P(A) + P(B)$ if event A and B are not disjointed, but the CEU model loosens this assumption. It states that under ambiguity, additivity is not necessary, and individuals are assumed to rely on capacities *v(.)* to evaluate events instead of probabilities (Schmeidler, 1989). I applied the power capacity function to capture how people distort their subjective probabilities, which are same as those in the EU model, in ambiguous situations: $v(p) = p^{\gamma}$, where $\gamma > 0$. In addition to be treated as a probability distortion parameter, γ reflects the degree of ambiguity aversion. If $\gamma > 1$, individuals would be ambiguity averse. If $1 > \gamma > 0$, they would be ambiguity lovers. Otherwise, this model collapses to the (S)EU model, and individuals would be seen as ambiguity neutral. This function is commonly used. Given a two-positive-outcome lottery, the power capacity will be identical to the probability weighting function applied in Cumulative Prospect Theory (Tversky & Kahneman, 1992). Also, if probabilities become objective, the CEU model with the power capacity can be directly treated as Rank-Dependent Utility Theory.

4.2 Hypothesis

Compared with the AMM and CEU models, the (S)EU model is the simplest. As the allocation problems become more complex, will the goodness-of-fit of the model change? While conventional wisdom suggests that complex problems require complex models (Bruhin et al., 2010; Conte & Hey, 2013), an opposite hypothesis is proposed in this project: as the complexity of ambiguous lotteries increases, simpler models like (S)EU may actually fit better.

This counterintuitive prediction follows Gabaix's idea (2014) that in complex environments, agents use simplified models (sparse maximization). If agents themselves simplify, a simpler structural model might better capture their actual decision process. Wilcox also argued that as complexity increases, the noise in subject responses increases, and complex models overfit increased decision noise (2011). From the cognitive overload prospective, several experimental evidences (Deck & Jahedi, 2015; Krajbich et al., 2009) also support this hypothesis. They stressed that when cognitive load is too high, individual choices become noisier and are better captured by simple utility models with added error terms rather than complicated models with more parameters.

Generally, complex models, like the AMM and CEU models, have more parameters. Estimating them requires more information. For instance, more choices and variations in the problem set. However, in complex tasks, cognitive overload may occur, leading to noisier choices from subjects. There is a risk that these complicated models, with their extra parameters, will simply overfit to this noise, mistaking it for a genuine behavioural pattern.

5. Econometric Method

As one of key points for this project is to check if individual attitudes toward ambiguity (or risk) would change as the complexity of the ambiguity increases estimating values of parameters in each part with different models mentioned above is crucial. The Maximum log-likelihood method is applied, and the list below (Table 1) summarizes parameters needed to be estimated. It should be stressed that estimation is conducted subject by subject.

A simplest risk-averse example is discussed here to demonstrate how to use model predictions and observations (which are discussed in Part 4) to infer the value of parameter *r*. In Part RL, the expected utility is $EU(X) = pu(XE_1, r) + (1-p)u((F-X)E_2, r)$. After differentiating it w.r.t. *X* and putting the answer equal to 0, $X = \frac{A}{A+B}F$, where $A = (1-p)^{-1/r}E_2^{1-1/r}$ and $B = p^{-1/r}E_1^{1-1/r}$. As values of *p*, E_1 and E_2 are known, the optimal *X* is a function of *r*, *X(r)*. Based on all observations and the optimal allocations given the value of parameters for each subject, the Maximum log-likelihood is used to find the best-fitting value of *r* subject by subject. A technical appendix is provided to check further details about optimal allocations under different models over parts. Risk seeking and risk neutral cases also are included in it.

Model	EU	RN[2]	AMM	CEU
Number of parameters	8	4	8[3]	8
Part RL	(r_{RL}, σ_{rRL})	σ_{rRL}	(r_{rRL}, σ_{rRL})	(r_{RL}, σ_{rRL})
Part ALA	(r_{ALA}, σ_{rALA})	σ_{rALA}	$(\alpha_{ALA}, \sigma_{\alpha ALA})$	$(\gamma_{ALA}, \sigma_{\gamma ALA})$
Part ALB	(r_{ALB}, σ_{rALB})	σ_{rALB}	$(\alpha_{ALB}, \sigma_{\alpha ALB})$	$(\gamma_{ALB}, \sigma_{\gamma ALB})$
Part ALC	(r_{ALC}, σ_{rALC})	σ_{rALC}	$(\alpha_{ALC}, \sigma_{\alpha ALc})$	$(\gamma_{ALC}, \sigma_{\gamma ALC})$

Table 1. Parameter estimation list

6. Results

The specific estimation results of parameters under different models are presented on the EXEC website. To test the hypothesis, the corrected Akaike Criterion (AICc) is applied, because the sample size for each participant in each part was different. AICc is computed based on the regular AIC which is calculated by *2k – 2Loglikelihood*, where *k* is the number of parameters of the model and *Loglikelihood* is the maximised log-likelihood[4]. With AIC, AICc is given by $AICc = AIC + \frac{2k(k+1)}{n-k-1}$, where *n* indicates the sample size. The lower the value of AICc, the corresponding model fits better. Figure 10 shows the fitness ranking results over parts[5]. Number '49' in Part ALA indicates that there are 49 participants' decisions can be explained by the EU model best, given 100 participants in total. Regardless of parts, the EU model seems to always be the best, although the CEU model also performed well in Part ALC. As for the AMM model which is more complicated than the EU model, the goodness-of-fit goes down as the allocation problems became more complex. Also, very few participants' behaviours are well captured by the risk-neutral (RN) case.

[2] RN indicates the risk neutral case based on the EU model (r = 0). It is considered as a benchmark when compare outcomes of the other three models.

[3] For the EU model, *r* was estimated in different parts with corresponding noise σ_r. As for the AMM model, α in each part was estimated based on the estimated value of *r* in Part RL. Similar logic is used to estimate γ in each part for the CEU model. If all three parameters r, α or γ and σ are estimated simultaneously, it will cause identifiability issues.

[4] The maximised log-likelihood in this project is the sum of the maximised log-likelihood of estimation in each part. For example, given the EU model, *Loglikelihood = llrRL+llrALA+llrALB+llrALC*, where *llrRL* represents the maximised loglikelihood of estimating *r* in Part RL, and *llrALA*, *llrALB* and *llr ALC* are named following the same logic to represent that in Part ALA, ALB and ALC respectively.

[5] As the complexity only changed in ambiguous lotteries, the comparison of model fitness focuses on Part ALA, ALB and ALC. The structure of ambiguous lotteries is more complex from ALA to ALC.

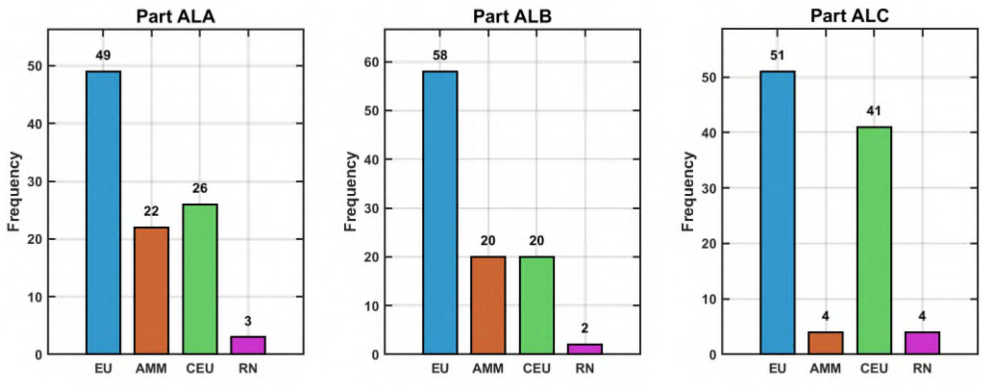

Figure 10. Distribution of the best fitting model over parts

To further check if the EU model significantly fits better than other two models, a series of one-tailed paired t-tests is used. To do this, we took the difference between the AICc of EU and the AICc of the other two models separately and tested whether they were significantly different from 0 against the alternative that the difference is negative. If the null hypothesis is rejected, the AICc for EU is significantly lower than the AICc of alternative models, and then it can be concluded that the EU model provides a significantly better fit. Table 2 displays test results for each part. The EU model does not significantly perform better than the AMM or CEU model in Part ALA, however, in Part ALB, where compared with Part ALA there was an extra decision stage, it fitted significantly better. As for Part ALC, consisted of a series of two-stage allocation problems with two ambiguous sub-lotteries, the EU model fits significantly better than the AMM model but not the CEU model. Such change in significance may be explained by that participants are more sensitive to the change in the number of stages. This pattern may be explained by participant adaptation to task complexity. In Part ALB, participants encountered a two-stage problem for the first time. The novel complexity likely induced cognitive overload, leading to noisier decisions that favoured simpler decision strategies (captured best by the parsimonious EU model). The more complex AMM model, with its additional parameters, was particularly prone to overfitting this high level of noise, explaining its poor fit throughout. By Part ALC, however, participants had become familiar with the two-stage structure. Their decisions were consequently less noisy, allowing the extra ambiguity-attitude parameter γ in the CEU model to reliably identify systematic behavioural patterns for a subset of participants, improving its fit relative to Part ALB.

Part	H_0	Mean Diff	Std Dev	Std Error[6]	t-Statistic	p-Value
ALA	AICcEU-AICcAMM=0	-5.356	37.886	3.789	-1.414	0.080
ALA	AICcEU-AICcCEU=0	-4.559	37.951	3.795	-1.201	0.116
ALB	AICcEU-AICcAMM=0	-13.305	30.900	3.090	-4.306	0.000***
ALB	AICcEU-AICcCEU=0	-12.132	28.519	2.852	-4.254	0.000***
ALC	AICcEU-AICcAMM=0	-40.904	134.627	13.463	-3.038	0.002**
ALC	AICcEU-AICcCEU=0	-8.853	98.550	9.855	-0.898	0.186

Table 2. AICc comparison results (Paired t-tests)

The observed pattern in parameter distributions supports this hypothesis. Distributions of the estimated *r* from the EU model are similar over all three parts involving ambiguous lotteries (Figure 11). As for distributions of estimated α (Figure 12) and γ (Figure 13), they are similar for the two more complex ambiguous allocation problems (in Part B and Part C) but differ for the simplest ones in Part ALA. The consistency of *r* suggests that risk attitude is a relatively stable trait that can be identified even in complex problems and well captured by the simplest EU model. α and γ, as the differentiating features of the more complex models, capture ambiguity attitudes, and well-identified in Part ALA. There is the consistency of estimated α and γ in this part, and both indicate that most participants tend to be ambiguity neutral (i.e. $\alpha = 0.5$ and $\gamma = 1$) given the simplest ambiguous problems. However, as problems become more complex (in Part ALB, ALC), the distributions of these parameters become similar but with extreme patterns[7]. This suggests that in complex choice environments, the behavioural 'signal' becomes too noisy to reliably estimate additional cognitive components. Consequently, the complex models lose their explanatory advantage and begin to overfit noise. On this side, the parsimonious EU model, which only estimates the stable risk attitude parameter *r*, provides a more robust and statistically superior account of the data, as it is not burdened by unreliable extra parameters.

[6] The difference of AICc is assumed to follow the normal distribution (0, σ^2), and the standard error is calculated by $SE = SD/\sqrt{n}$, where n = 100 indicating the sample size.

[7] For example, estimated α concentrated on 0 or 1 which means fully pessimistic or optimistic for individuals toward ambiguity. As for γ in the CEU model, a significant portion of estimated value is just slightly above 0 which indicates an extreme ambiguity seeking attitude.

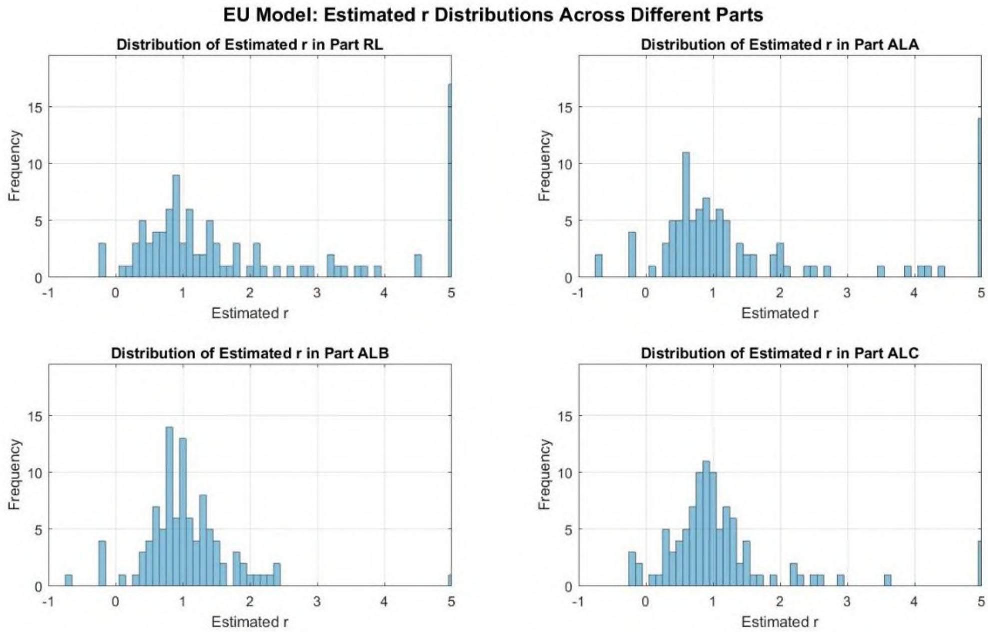

Figure 11. Distribution of estimated r across parts

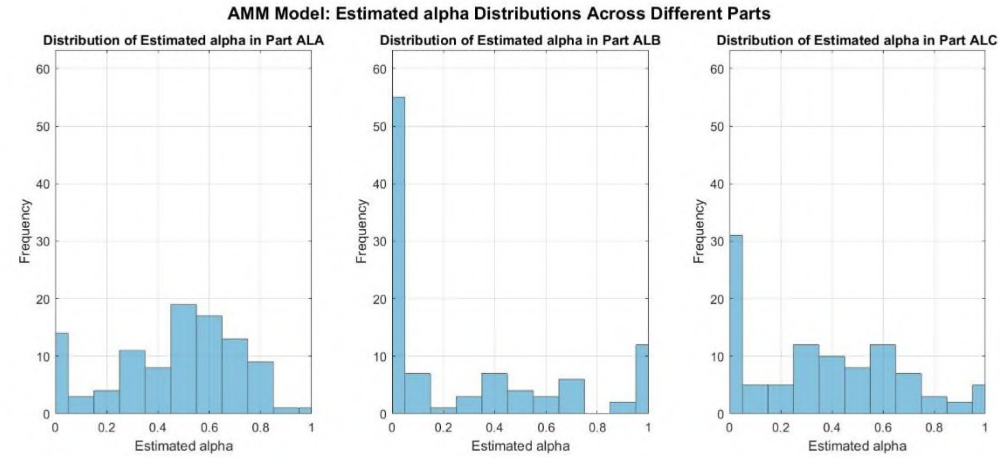

Figure 12. Distribution of estimated α across parts

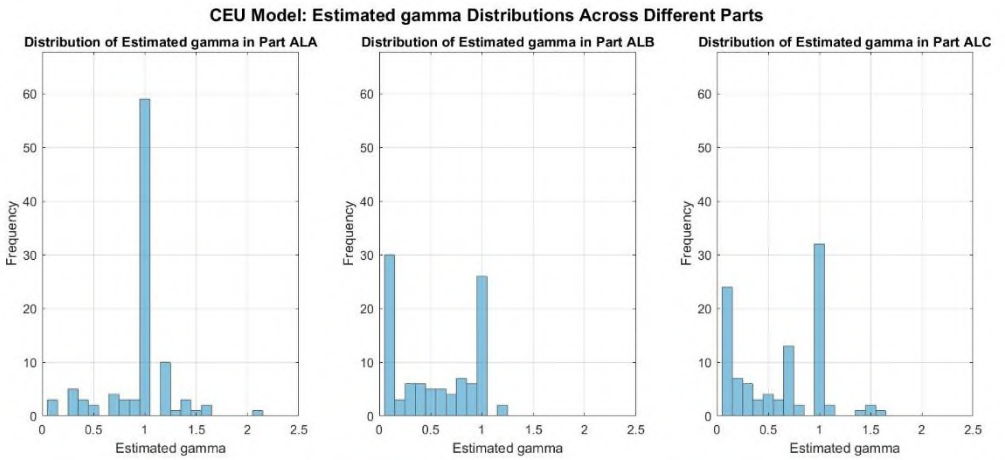

Figure 13. Distribution of estimated γ across parts

A series of t-tests about estimated α and γ also supports above explanation. In Part ALA, both null hypotheses α =0.5 and γ=1 were failed to be rejected separately but not in Part ALB and ALC (details see Table 3 and Table 4).

Test	Part	Mean α	Std α	t-statistic	p-value
H_0: α =0; H_1: α >0	ALA	0.451	0.254	17.782	0.000^{***}
H_0: α =0.5; H_1: $\alpha \neq$0.5	ALA	0.451	0.254	-1.936	0.056
H_0: α =1; H_1: α <1	ALA	0.451	0.254	-21.654	0.000^{***}
H_0: α =0; H_1: α >0	ALB	0.262	0.360	7.295	0.000^{***}
H_0: α =0.5; H_1: $\alpha \neq$0.5	ALB	0.262	0.360	-6.602	0.000^{***}
H_0: α =1; H_1: α <1	ALB	0.262	0.360	-20.498	0.000^{***}
H_0: α =0; H_1: α >0	ALC	0.344	0.308	11.154	0.000^{***}
H_0: α =0.5; H_1: $\alpha \neq$0.5	ALC	0.344	0.308	-5.055	0.000^{***}
H_0: α =1; H_1: α <1	ALC	0.344	0.308	-21.263	0.000^{***}

Table 3. t-test results of estimated α

Test	Part	Mean γ	Std γ	t-statistic	p-value
H_0: γ=1; H_1: $\gamma \neq$1	ALA	0.953	0.320	-1.461	0.147
H_0: γ=1; H_1: $\gamma \neq$1	ALB	0.551	0.379	-11.861	0.000^{***}
H_0: γ=1; H_1: $\gamma \neq$1	ALC	0.618	0.410	-9.317	0.000^{***}

Table 4. t-test results of estimated γ

Such series of finding discloses that the EU fits best relatively, and the absolute goodness-of-fit of EU should be further investigated. To evaluate it, we computed McFadden's pseudo-R^2 ($=1-\frac{\ln L_{full}}{\ln L_{null}}$), where $\ln L_{full}$ and $\ln L_{null}$ denote log-likelihoods of the fitted model and the null model respectively[8]. This indicator quantifies the improvement in log-likelihood of the fitted

[8] $\ln L_{full}$ and $\ln L_{null}$ are calculated in the aggregate level, the sum of loglikelihood of all observations.

model relative to a null model. In our analysis, we employ the risk-neutral case (r = 0) as the null. Values of pseudo-R^2 between 0.2 and 0.4 are generally considered to indicate an excellent fit in discrete choice models (McFadden, 1979). Table 5 shows the results of pseudo-R^2 over Part ALA, ALB and ALC, and discloses that the EU model absolutely fits data well cross these parts.

Part	LL-EU	LL-RN	Pseudo-R^2
ALA	-4044.736	-14404.821	0.719
ALB	-11209.791	-17210.016	0.349
ALC	-16265.526	-22440.483	0.275

Table 5. Pseudo-R^2 results

7. Conclusion

This research employed the allocation method in an experimental setting to investigate how individuals' decision-making adapts as choice problems grow in complexity, and to evaluate which theoretical model (the Expected Utility (EU) model, the αMaxMin (AMM) model, or the Choquet Expected Utility (CEU) model) best captures these behavioural patterns under ambiguity. Complexity was reflected in two key dimensions: the number of possible outcomes and the number of decision stages.

The results provide strong empirical confirmation of our core hypothesis: increasing complexity under ambiguity favours simpler models. As allocation problems grew more complex, particularly with the novel introduction of multiple stages, the EU model consistently provided the most robust and statistically superior fit. This finding directly challenges the conventional wisdom that complex problems necessitate complex models. Instead, it demonstrates that cognitive overload in complex environments leads to noisier choices, which causes complex models to overfit decision noise while simpler models remain robust. In essence, increasing complexity incentivizes cognitive simplification by decision-makers (DMs), a reality best captured by a simpler structural model.

The AICc comparisons and the distribution of estimated parameters jointly support this conclusion. The EU model's single risk-attitude parameter r proved stable and reliably identifiable across all tasks, underscoring that core risk preference is a resilient trait even under cognitive strain. In stark contrast, the more complex AMM model, with its greater parametric demands, consistently underperformed; its additional parameters became liabilities, overfitting the increased decision noise. The CEU model's performance was more nuanced. It regained explanatory power only after participants acclimated to the two-stage format, suggesting that a reduction in cognitive noise can allow a well-specified extra parameter γ, which captures ambiguity sensitivity, to identify meaningful behaviour. This pattern illustrates the critical tension between model complexity and data fidelity, validating the theoretical frameworks of sparse maximization and cognitive overload.

These results carry significant implications for economic modelling and real-world decision architecture. They demonstrate that the 'best' model of human choice is not absolute but context-dependent, varying with the cognitive demands of the decision environment. In real-world ambiguous settings characterized by novelty, time pressure, or information overload, such as rapid financial trading, complex insurance choices, or emergency response planning, individuals likely default to simpler, EU-like heuristic strategies. Policymakers and designers should therefore favour robust, parsimonious models to avoid the pitfall of overfitting to transient noise. Conversely, in familiar, structured, and low-pressure contexts (e.g., long-term pension planning), richer models like CEU may become applicable as decision noise subsides.

Ultimately, this study argues for a ‘cognitive complexity calibration’ in applied economics: matching the complexity of our behavioural models to the cognitive reality faced by the decision-maker leads to more accurate predictions and more effective interventions. Our findings affirm that in a world of bounded rationality, especially under the joint pressure of ambiguity and complexity, simplicity is not merely a modelling virtue—it is a fundamental behavioural reality.

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